Math 242, S13 Name: Answer
Quiz 11 SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Use Laplace transform to solve the initial value problem:

$$x'' + 16x = 0; \quad x(0) = 4,\ x'(0) = 5.$$

Solution: let $X(s) = \mathcal{L}\{x(t)\}$, then

$$\mathcal{L}\{x''(t)\} = s^2X(s) - sx(0) - x'(0) = s^2X(s) - 4s - 5$$

$$\Rightarrow \mathcal{L}\{x'' + 16x\} = \mathcal{L}\{0\}$$

$$\mathcal{L}\{x''\} + 16\mathcal{L}\{x\} = 0$$

$$s^2X(s) - 4s - 5 + 16X(s) = 0$$

$$(s^2+16)X(s) = 4s+5$$

$$X(s) = \frac{4s+5}{s^2+16}$$

$$\Rightarrow x(t) = \mathcal{L}^{-1}\left\{\frac{4s+5}{s^2+16}\right\} = 4\mathcal{L}^{-1}\left\{\frac{s}{s^2+16}\right\} + 5\mathcal{L}^{-1}\left\{\frac{1}{s^2+16}\right\}$$

$$= 4\cos 4t + \frac{5}{4}\sin 4t$$

Problem 2. (4 points) Find the inverse Laplace transform of $\dfrac{1}{s(s-4)}$ using the formula $\mathcal{L}^{-1}\left(\frac{F(s)}{s}\right) = \int_0^t \mathcal{L}^{-1}\{F(s)\}d\tau$.

Solution: $\mathcal{L}^{-1}\left\{\frac{1}{s(s-4)}\right\} \overset{F(s)=\frac{1}{s-4}}{=\!=} \mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t \mathcal{L}^{-1}\left\{\frac{1}{s-4}\right\} d\tau$

$$= \int_0^t e^{4\tau} d\tau$$

$$= \left[\frac{1}{4}e^{4\tau}\right]_{\tau=0}^{\tau=t}$$

$$= \frac{1}{4}e^{4t} - \frac{1}{4}$$

$$= \frac{1}{4}(e^{4t} - 1)$$